Devender Singh

devenders@mun.ca · idevender.github.io · github.com/idevender · Google Scholar

Education

Memorial University of Newfoundland May 2022 – May 2027 (expected)
B.Sc. (Honours), Computer Science · GPA 3.94/4.00

- Honours thesis, *Terrain-Aware Local Path Planning with Global DEM Data Integration for Autonomous UGV Navigation* (94/100), supervised by Vinicius Prado da Fonseca and Matthew Hamilton, published as [3] below.
- Mathematics coursework in Linear Algebra, Calculus I–III, Ordinary Differential Equations, Statistics, and Real Analysis (Fall 2026).

Publications & Preprints

The Loss Does Not See the Basis, but Adam Does 2026
Devender Singh (sole author)
Preprint, arXiv:2608.05136. Code at github.com/idevender/loss-basis-adam

FlowAdam: Implicit Regularization via Geometry-Aware Soft Momentum Injection 2026
Devender Singh, Tarun Sheel
IJCNN 2026, IEEE World Congress on Computational Intelligence, Maastricht. Oral presentation. arXiv:2604.06652

Terrain-Aware Local Path Planning with Global DEM Data Integration for Autonomous UGV Navigation 2026
Devender Singh, Issah Nazif Suleiman, Paul Mitten, Glenn Cutler, Vinicius Prado da Fonseca, Matthew Hamilton
Preprint, arXiv:2608.17038

Improved Simulated Viewing of Holographic Imagery 2023
Nicholas Wells, **Devender Singh**, Haoran Wang, Matthew Hamilton, Oscar Meruvia-Pastor, Amilcar Soares
32nd IEEE Newfoundland Electrical and Computer Engineering Conference (NECEC), non-archival

Research Experience

Research Assistant (MUCEP), Mathematics and Statistics, Memorial University Fall 2023 – present
Consecutive research terms with Tarun Sheel on the geometry of adaptive optimizers, which produced FlowAdam [2]. The work in [1] is my own. It shows that gradient descent's low-rank implicit bias transfers only to optimizers that respect the gauge symmetry of the loss, and measures where Adam does not, in matrix sensing, transformers, LoRA adapters and model merging.

Mitacs Business Strategy Internship, VACLab, Memorial University Summer 2024 (6 months)
Terrain-aware local path planning for autonomous ground vehicles, in collaboration with Compusult Ltd. Continued as the Honours thesis and published as [3].

Summer Undergraduate Research Award (SURA), VACLab, Memorial University Summer 2023
Simulation and rendering for light-field displays, advised by Matthew Hamilton. The resulting method was presented at NECEC 2023 and appears as [4].

Industry Experience

AI Engineer, Verlo (AI startup, St. John's) · **12-month co-operative work term** May 2025 – April 2026
Rebuilt the retrieval and agent pipeline behind an AI-native wealth platform, isolating tool failures from the response stream, pruning the token budget, and replacing a hosted vision model with local document parsing. Inference cost fell 70% and reliability rose to 93.8% over a corpus of 1.2M documents.

Honours & Awards

- INNS Travel Grant (International Neural Network Society), with travel and engagement awards from the Office of the AVP Academic and Dean of Students, the Dean of Science, the City of St. John's, and MUNSU 2026
- Mitacs Business Strategy Internship Award 2024
- Summer Undergraduate Research Award (SURA), Faculty of Science, Memorial University 2023
- Dean's List, Faculty of Science, Memorial University